

Here are five MCQs based on the context provided:

Q1: What is the mathematical form of Newton's 2nd law of motion?

- A) $F = m / a$
- B) $F = m * a$
- C) $F = a / m$
- D) $F = m + a$

Answer: B) $F = m * a$

Explanation: The context directly states that $F = m * a$ is the mathematical form of Newton's 2nd law of motion.

Q2: According to Newton's 1st law of motion, what happens to a body at rest if no net force acts on it?

- A) It starts moving with uniform speed
- B) It remains at rest
- C) It starts moving with accelerated motion
- D) It changes its direction

Answer: B) It remains at rest

Explanation: The context states that a body at rest will remain at rest if no net force acts on it, as per Newton's 1st law of motion.

Q3: What is the relationship between force, mass, and acceleration according to Newton's 2nd law of motion?

- A) Force is directly proportional to mass and inversely proportional to acceleration

- B) Force is directly proportional to acceleration and inversely proportional to mass
- C) Force is directly proportional to mass and acceleration
- D) Force is independent of mass and acceleration

Answer: B) Force is directly proportional to acceleration and inversely proportional to mass

Explanation: The context explains that acceleration is directly proportional to the applied force and inversely proportional to the mass of the body.

Q4: (Application) A car of mass 1000 kg is moving with a uniform speed of 50 m/s. What is the net force acting on the car?

- A) 5000 N
- B) 0 N
- C) 1000 N
- D) 2000 N

Answer: B) 0 N

Explanation: According to Newton's 1st law of motion, if the car is moving with uniform speed, the net force acting on it is zero.

Q5: (Analysis) Two objects, A and B, interact with each other. Object A exerts a force of 10 N on object B. What is the force exerted by object B on object A, according to Newton's 3rd law?

- A) 5 N in the same direction
- B) 10 N in the opposite direction
- C) 5 N in the opposite direction
- D) 20 N in the same direction

Answer: B) 10 N in the opposite direction

Explanation: Newton's 3rd law states that every action has an equal and opposite reaction, so the force exerted by object B on object A is equal in magnitude and opposite in direction to the force exerted by object A on object B.